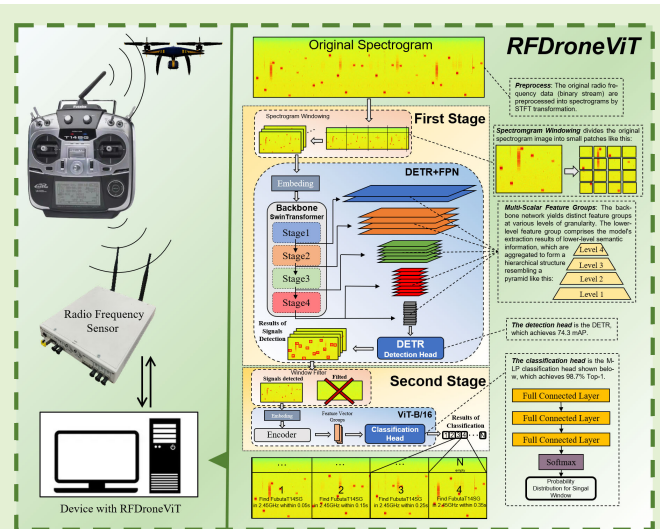


A Radio Frequency Sensor-Based UAV Detection and Identification System Using Improved Vision Transformer-Based Model

Lu Xu, Rui Shi, and Yijia Zhang

Abstract—In this article, we propose a novel approach to addressing the challenge of unmanned aerial vehicle (UAV) detection and identification by combining object detection and image classification techniques. Diverging from prevailing research that predominantly relies on convolutional neural networks (CNNs), we introduce the radio frequency drone vision transformer (RFDroneViT), a novel two-stage vision transformer (ViT) consisting of two distinct sub-ViT. The first stage employs the designed end-to-end object detection with transformer (DETR) to detect the UAV, while the second stage utilizes ViT-B/16 to identify the detected UAV. Based on RFDroneViT, we propose several detailed optimization algorithms, culminating in the development of a cost-effective system capable of UAV detection and identification. We evaluate the system's performance through rigorous experimentation. Our custom DETR model achieves the state-of-the-art performance in drone signal detection with an average precision (AP) of 74.3 and demonstrates an 8.7 improvement in small target AP (AP_S) for the signal detection task compared to the original DETR. Additionally, the classification model can achieve 98.7% Top-1. The experimental results demonstrated that our proposed system achieves great performance on drone detection and identification.

Index Terms—Improved designed end-to-end object detection with transformer (DETR)-based model, radio frequency (RF)-based anti-unmanned aerial vehicle (UAV) system, spectrogram sensing technology, vision transformer (ViT).



I. INTRODUCTION

IN RECENT years, the rapid advancement of unmanned aerial vehicle (UAV) technology has precipitated a surge in the popularity of UAV-related products in the market. The global drone market is projected to achieve a valuation of U.S. \$55.8 billion by 2030 [1]. The rapid advancement of UAVs has greatly facilitated various aspects of our daily lives, while concurrently sparking public concerns regarding privacy and security, thereby presenting substantial regulatory challenges [2].

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Numerous researchers have designed various anti-UAV systems targeting different characteristics of drones, such as appearance, sound, and communication methods. These systems include radar detection system, acoustics detection system, visual detection system, and radio frequency sensor-based (RF-based) detection system.

The radar detection system exploits the backscattered RF signals to detect and identify UAVs. The boosting classifier outperforms the support vector machine (SVM) in distinguishing drones from birds and identifying drone types in the micro-Doppler radar [3]. In millimeter-wave (mm-wave) frequency-modulated continuous-wave (FMCW) radar, high accuracy can be achieved using various machine learning models, and the custom lightweight convolutional neural network (CNN) is demonstrated that well-suited for embedded or edge computing devices [4]. Although the radar system performs high accuracy in both detection and identification, its exorbitant manufacturing cost renders it unsuitable for civilian applications.

The acoustics detection system detects the noise generated by flying UAVs. The researchers have successfully validated

a time–frequency processing technique that leverages the spectral and spatial attributes of ego noise [5], as well as a Bayesian-filter-based time difference of arrival (TDOA) estimation algorithm [6]. The acoustics detection system is susceptible to interference from ambient noises, thereby rendering it unsuitable for environments with high noise levels such as urban areas. Compared with these methods, our method is not affected by most natural environmental factors.

The visual detection system utilizes cameras to capture UAVs. The CNN-based image recognition algorithm demonstrates excellent performance [7], [8]. The effectiveness of this system is contingent upon the clarity of images captured by cameras, rendering it vulnerable to interference caused by adverse weather conditions. In contrast to these approaches, our method remains unaffected by unforeseen circumstances and exhibits applicability across a broader spectrum of scenarios.

The RF-based detection system leverages the inherent characteristic of drones maintaining continuous communication with their controllers throughout the flight. These methods use a receiver for signal acquisition within the industrial scientific medical (ISM) bands (985 MHz, 2.4 GHz, and 5.8 GHz) commonly employed in drone-controller communication [9]. The collected data undergo a series of signal-processing techniques to detect the drone. Compared with methods aforementioned, the RF-based detection system offers cost-effectiveness, ease of deployment, and robust detection performance. Thus, it is well-suited for diverse environments.

Conventional RF-based methodologies necessitate the decoding of data packets transmitted between the drone and controller. The data content, consisting of flight paths and drone-type information from dajiang innovations (DJI) drone signals, was successfully unpacked using orthogonal frequency-division multiplexing (OFDM) decoding techniques [10]. These approaches heavily rely on communication protocols, and the entire procedure necessitates manual screening.

The application of deep learning in RF analysis obviates the need for demodulation and decoding. A passive drone detection and classification system is proposed in [11], including K-nearest-neighbor (KNN), discriminant analysis (DA), SVM, and neural networks (NN). This method heavily relies on prior knowledge, rendering the entire system susceptible to potential vulnerabilities. In contrast, our model offers an end-to-end detection method that exhibits enhanced robustness. A CNN-based approach is proposed for the simultaneous detection and identification of UAVs, employing the you only look once (YOLO [12]) algorithm for drone detection and utilizing deep residual neural networks (DRNNs) to identify the detected drones [13]. Its performance surpasses that of goodness-of-fit (GoF) spectrum sensing technology. Another proposed model, a time–frequency multiscale CNN-based approach, effectively captures features from both the time domains and frequency domains, thereby exhibiting superior performance on a publicly available dataset [14]. These CNNs only identify a limited number of drone types. Additional validation is necessary for a model initially designed to detect and classify numerous categories. In contrast, our model demonstrates a significantly

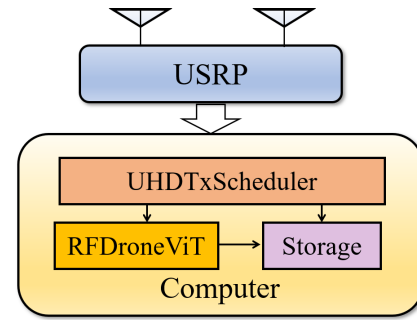


Fig. 1. Simulated physical system structure diagram. The UHDTxScheduler is a component and functionality within the USRP hardware driver (UHD) and SDR environment, specifically focused on scheduling and managing the transmission of data from the host to the USRP device.

enhanced capability in recognizing various types of drones, achieving an approximate fivefold increase. Moreover, our experiment incorporates more precise model metrics and provides comprehensive data for comparing different models.

In order to enhance the detection performance, we have implemented various improvements in the model structure and proposed several novel algorithms. The main contributions of this article are as follows.

- 1) We introduce a two-stage vision transformer (ViT) into the task of UAV detection and identification and propose a comprehensive system. In comparison to other RF-based drone detection and classification algorithms, our approach significantly minimizes the need for manual intervention, eliminating the requirement of having a signal processing expert available round-the-clock at a receiving station.
- 2) To address the challenge of detecting small and densely packed drone signals in spectrograms, we incorporate the feature pyramid network (FPN) [15] into the designed end-to-end object detection with transformer (DETR) framework [16].
- 3) To enable the first-stage model to detect drone signals across a wide spectrum, we design the *spectrogram windowing algorithm* that effectively reduces the inference time complexity of the model on a high-resolution spectrogram.
- 4) To enhance the performance of the classification model, we design the *spectrogram window filtering algorithm* to reduce the missing rate.

II. SYSTEM MODEL

Section II will encompass: 1) the structural framework of the system; 2) the difficulty of achieving the task and preprocessing techniques applied to the raw data acquired by the receiver; and 3) a practical experiment demonstrating the efficacy of the system.

The architecture of our system, consisting of a processor and a receiver, is depicted in Fig. 1. The processor can refer to any computational and storage device, such as a computer, edge device, or embedded system. The receiver can be any software-defined radio (SDR) system, including universal software radio peripheral (USRP) or blade RF.

In this article, we choose a high-performance computational server equipped with the radio frequency drone vision transformer (RFDroneViT) as the designated processor. The receiver is a USRP due to its renowned versatility and reliability. The RF data collected by the USRP are stored as binary data streams, ensuring efficient and accurate data retention.

The utilization of an RF-based method for drone detection relies on the identification of patterns within the amplitude fluctuations present in the raw signal data. The direct identification of these implicit patterns poses significant challenges. Hence, the explicit mapping of features derived from the raw data plays a pivotal role in determining the system's implementation complexity. The raw signal is subjected to a short-time Fourier transform (STFT) in this study. The following equation shows the STFT transformation:

$$\text{STFT}(X) = \int_{-\infty}^{\infty} x(\tau) \cdot w(t - \tau) \cdot e^{j2\pi f\tau} d\tau. \quad (1)$$

Here, $x(\tau)$ presents the raw signal data, $w(\tau)$ presents the window function, and f presents the frequency. The STFT output is visually depicted through a color-coded representation.

By transforming the binary data stream into the time-frequency domain, we can effectively capture its dynamic characteristics, thereby facilitating detection and classification with enhanced accuracy and robustness. A portion of the spectrogram results in the dataset are presented in Appendix I.

As evident from these spectrograms, the inherent characteristics in the raw data exhibit variations attributed to diversity due to different communication protocols employed between the drone and its controller. For example, DJI's second-generation communication protocol, BigZee, supports Phantom, Matrice, and Inspire series drones. The following equation is the modulation of the uplink control signals, which utilizes frequency hopping spread spectrum (FHSS) technology [17]:

$$f(t) = A \sum_{-\infty}^{+\infty} W_T(t - kT_h) \cos[2\pi f_k(t - kT_h) + \phi_n]. \quad (2)$$

Here, W_T represents a rectangular window function with a width of T_h , A denotes the modulation amplitude, $f_0, f_1, \dots, f_k$ ($k = 0, 1, 2, \dots, N - 1$) is a set of equidistant frequency points, and ϕ_n ($n = 0, 1, 2, \dots, N - 1$) is a set of equidistant initial phase values. Equation (2) indicates that different types of drones exhibit variations in their frequency points and the spacing between these points. Furthermore, the modulation method of frequency hopping (FH) signals is fixed after the drones are manufactured. Formally, this approach is also analogous to directly recognizing drone fingerprints [18].

The drone detection method proposed in this article primarily operates by identifying various drone signals present within the spectrum. These signals can be distinguished based on their individual transmission durations and bandwidths. Our research and experimentation have revealed that certain video transmission signals typically exhibit a duration of approximately 0.01 s and a bandwidth of around 20 MHz. Due to their relatively large presence in the spectrogram, these signals are more easily detectable for models employed in this

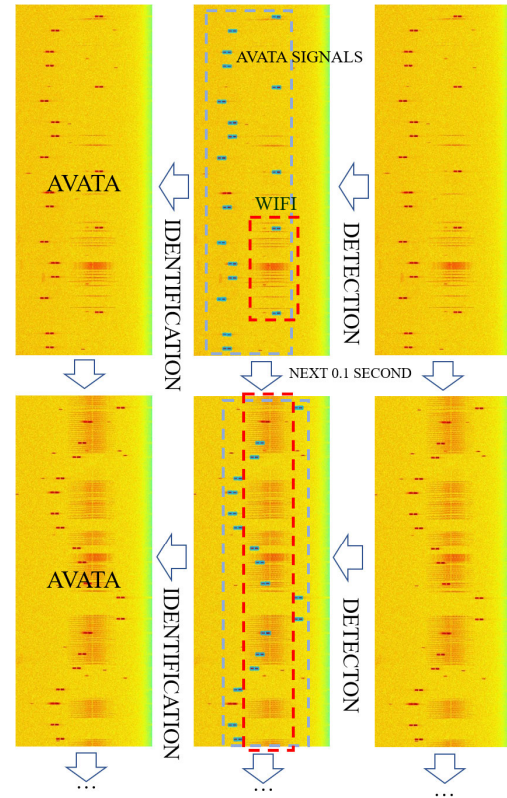


Fig. 2. Results of the system detecting and classifying the AVATA drone in the presence of Wi-Fi signal interference.

study. For this task, we refer to these signals as “conventional signals.”

Conversely, a multitude of frequency-hopping signals, characterized by a duration of approximately 0.001 s and a bandwidth of about 2 MHz, exhibit condensed and frequent appearances within the spectrogram while occupying a smaller spatial extent. In this study, we classify these signals as “small target signals.”

Our experimental findings indicate that detecting such diminutive signals presents greater challenges for model detection compared to conventional signals, thereby resulting in an elevated probability of missed detections.

The detection of small targets has long posed a formidable challenge in traditional object detection tasks. Prominent strategies to tackle this issue encompass enhancing feature maps for small objects [19], [20], incorporating contextual information pertaining to small objects [21], addressing the imbalance between foreground and background classes for small objects [22], and augmenting training examples specifically tailored for small objects [23].

In this article, we employ a multilevel feature map fusion approach derived from the FPN to enhance the model's ability to detect small target signals, effectively solving this challenge.

Building on the aforementioned data preprocessing methods, we conduct a straightforward experiment to showcase our system's effectiveness. The processor and receiver are situated in a room where they must contend with interference from other signals, such as Wi-Fi and Bluetooth. The target of this experiment involves an AVATA drone, which is maneuvered in

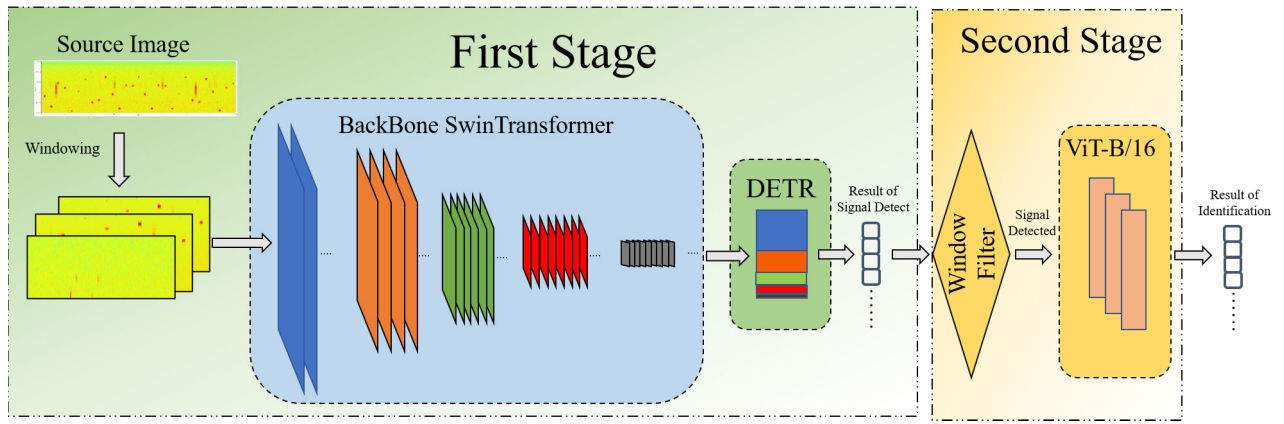


Fig. 3. Overview of the RFDroneViT model. The colorful rectangle refers to the data flow in each part of the model.

a reciprocating motion close to the receiver. Fig. 2 shows the processing results at each stage. The results of two consecutive signal segments illustrate that despite the presence of noise interference, the system successfully detects and identifies the target. Additionally, we will conduct a series of comprehensive experiments to further demonstrate the robustness of our system as elaborated in Section IV-C.

III. RF DRONE VISION TRANSFORMER

Section III will encompass: 1) the structure and forward propagation of improved DETR and 2) system implementation details. Mathematical expression combined with the model structure diagram will be used to present how to introduce the FPN structure backbone into DETR. The formulation presented in this article is partially inspired by the structural expression employed in Deformable-DETR [24]. Therefore, if one lacks familiarity with the attention mechanism or the fundamental working principles of DETR, comprehending these mathematical expressions may prove challenging. Additional explanations pertaining to the mathematical formulas utilized in Deformable-DETR, DETR, and attention mechanisms can also be found [25].

Following the aforementioned preprocessing methods, we obtain a spectrogram containing drone signals. This spectrogram is subsequently processed by the RFDroneViT.

The structure of RFDroneViT is illustrated in Fig. 3. The first stage of RFDroneViT encompasses *spectrogram windowing*, where the entire spectrogram is divided into several subspectrograms of identical resolution, and *signal detection*, wherein these subspectrograms are sequentially fed into an improved DETR model for the detection of drone signals.

The second stage focuses on the comprehensive analysis of the detected signals and includes *spectrogram window filtering*, which involves filtering out empty subspectrograms and fine-tuning those containing detected signals for classification purposes, and *spectrogram window classification*, where the filtered subspectrograms are input into the ViT-B/16 model to determine the specific drone type.

A. First Stage: Signal Detection Based on Improved DETR Model

Fundamentally, the first-stage model involves an object detection task within the field of computer vision. But why

TABLE I
RESULT OF THE FIRST-STAGE MODEL COMPARISON

Model	AP	AP ₇₅	AP ₅₀	AP _S	AP _M	AP _L
YOLOV5	67.1	79.6	93.5	43.8	69.1	87.9
YOLOV8	68.1	80.1	94.1	44.8	69.7	88.9
Improved YOLOV8	68.9	80.9	94.3	49.2	69.6	88.8
Faster-RCNN(R50)	58.9	67.9	81.6	37.4	63.8	77.9
Faster-RCNN(R101)	59.3	68.1	82.1	38.9	64.5	78.7
Faster-RCNN+FPN(R50)	64.5	73.1	86.2	42.7	68.1	80.5
Faster-RCNN+FPN(R101)	65.1	73.6	86.8	42.8	66.7	81.9
DAB-DETR(R50)-DC5	67.9	75.7	95.6	47.4	67.9	78.6
Deformable-DETR(R50)	68.6	75.7	95.6	47.4	67.9	78.6
DINO-5scale(R50)	70.4	78.5	95.8	52.4	68.5	79.4
DETR+SwIn-T+FPN(ours)	72.4	82.1	95.9	58.9	69.7	81.8
DETR+SwIn-B+FPN(ours)	73.9	83.1	96.4	60.4	70.4	82.2
DETR+SwIn-L+FPN(ours)	74.3	84.6	98.9	62.9	71.8	84.8

choose DETR among the plethora of excellent object detection networks? This choice is validated by our experimental results, as detailed in Table I.

Theoretically, the effectiveness of this approach is intrinsically linked to the architecture of DETR. DETR employs ViT [26] for object detection, functioning as a comprehensive end-to-end detector. In contrast to traditional two-stage networks such as Faster-RCNN [27], DETR provides enhanced consistency in processing. Lightweight networks like YOLO encounter difficulties in detecting multiple signal targets within a single image, often leading to missed detections. The deeper architecture of DETR facilitates the effective handling of complex patterns and variations in signal detection, thereby mitigating this problem.

Furthermore, the end-to-end design of DETR eliminates the need for hand-crafted anchor boxes and nonmaximum suppression, simplifying the detection pipeline and improving robustness. This structural advantage, combined with its ability to learn global context through self-attention mechanisms, enhances its performance in diverse and challenging environments.

However, DETR exhibits suboptimal performance in recognizing small-signal targets. The primary reason stems from its reliance on the backbone for feature extraction. The inefficiency of the backbone network to extract features from input images has a detrimental impact on the detection of small targets. Inspired by FPN and other DETR-like models, including DETR with improved denoising anchor boxes for end-to-end object detection (DINO) [28], DETRs with

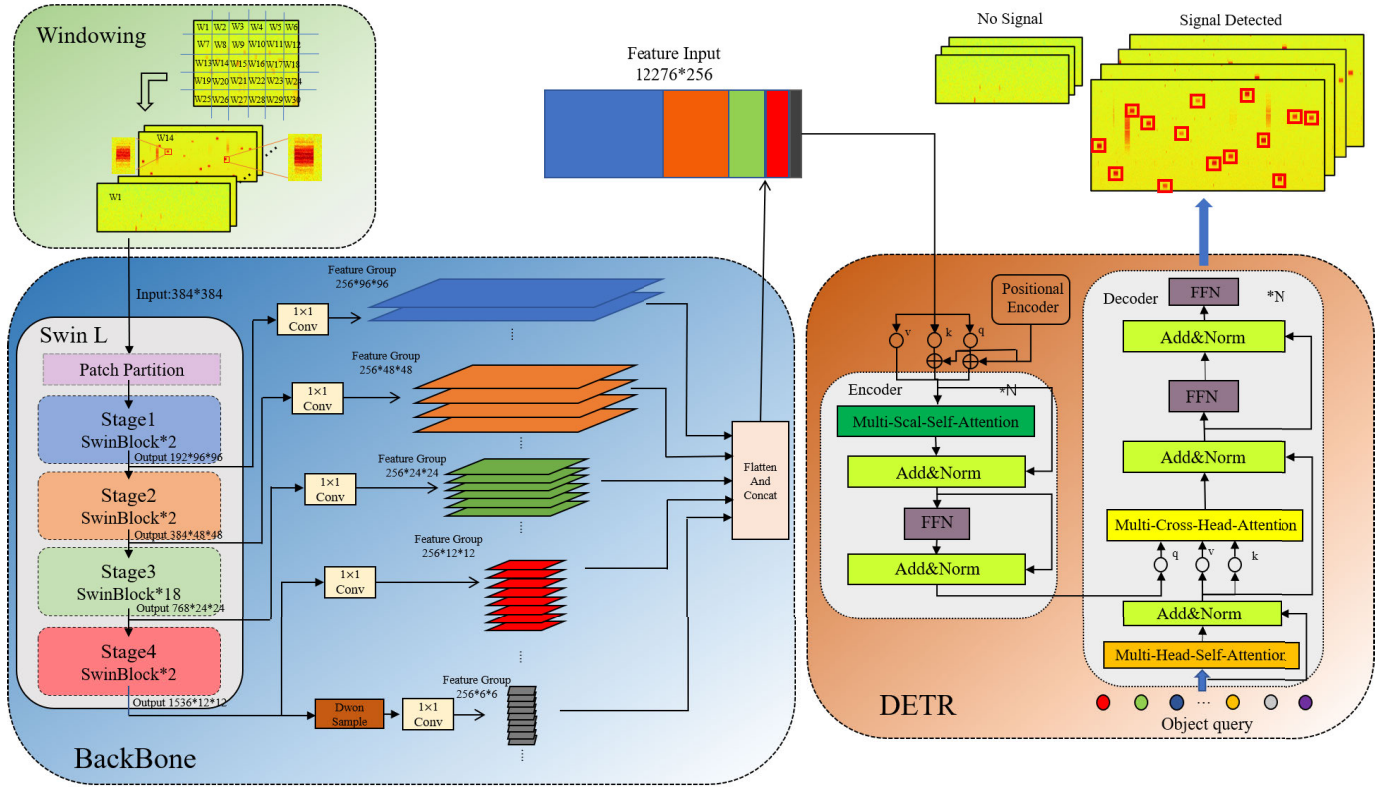


Fig. 4. Structure of the first-stage model (improved DETR). The enhancements primarily focus on the backbone module. The output of each stage in the Swin Transformer is utilized as a new feature scale (the stacked colorful rectangle). The 1×1 Conv is employed to standardize the feature scales, ensuring uniformity in shape, and subsequently flattening and concatenating them into a comprehensive feature vector (feature input). This vector is then fed into DETR for further processing.

collaborative hybrid assignments training (CO-DETR) [29], Deformable-DETR, and dynamic anchor boxes are better queries for DETR (DAB-DETR) [30], we seamlessly integrated this fine-grained feature extraction method into DETR's backbone. During the training phase, we employed parallel auxiliary heads supervised by one-to-many label assignments, such as Faster-RCNN, to facilitate enhanced convergence of DETR.

FPN proposes the merging of feature maps from various stages of backbone processing into a new channel, resulting in feature maps with enriched semantic characteristics, while preserving detailed features. Building on this concept, we propose an enhanced network that achieves superior detection performance. Specifically, we have integrated FPN with the Swin Transformer [31] as the backbone for feature extraction from raw data. The architecture of the entire network is depicted in Fig. 4.

Initially, an subspectrogram $X \in \mathbb{R}^{3 \times H \times W}$ is fed into the Swin Transformer. Outputs from each stage of the Swin Transformer are treated as feature maps at different scales of subspectrogram, denoted as $\{X_l^n\}_{l=1}^L$ (n denotes the n th swin-block, and l denotes the number of stages, $L = 5$).

The following equations illustrate the calculation of the 1th to fourth feature map at the l th scale:

$$X_l^n = W_l^n \cdot \text{LN}[\text{SW-MSA}\{\text{LN}(\hat{X}_l^n)\}] \quad (3)$$

$$\hat{X}_l^n = \hat{W}_l^n \cdot \text{LN}[\text{W-MSA}\{\text{LN}(X_l^{n-1})\}]. \quad (4)$$

Here, $\hat{X}_l^n$ represents the transitional state between the encoder and decoder, W_l^n represents the weights of the fully connected (FC) layer connected after the n th encoder in the l th stage, and $\hat{W}_l^n$ represents the weights of the FC layer connected after the n th decoder in the l th stage. LN denotes the layer normalization [32] function, as shown in (7), W-MSA represents the windowed multihead self-attention function, as shown in (5), and SW-MSA represents the shifted windowed multihead self-attention function as shown in (6)

$$\text{SW-MSA}(X) = \sum_{m=1}^M W_{\text{SW-MSA}} \left[\frac{(x_m^T x_m)}{\sqrt{d}} x_{m+\Delta v} \right] \quad (5)$$

$$\text{W-MSA}(X) = \sum_{m=1}^M W_{\text{W-MSA}} \left[\frac{(x_m^T x_m)}{\sqrt{d}} x_m \right] \quad (6)$$

$$\text{LN}(x) = \frac{(x - \mu)}{\sqrt{\sigma + \epsilon}} \cdot \gamma + \beta. \quad (7)$$

In (5) and (6), M represents the number of attention heads, X indicates the current input state, and Δv signifies the window offset. $W_{\text{W/SW-MSA}}$ represents the weights of the FC layer connected after the windows/shifted windows attention block. In (7), μ is the mean of the current layer, σ is the variance of the current layer, and ϵ is a positive number. Both γ and β are the hyperparameters.

After processing through the four stages of the Swin Transformer, we obtain feature maps from the original

subspectrogram with dimensions

$$\begin{aligned} \mathbf{x}_1 &\in \mathbb{R}^{192 \times \frac{H}{4} \times \frac{W}{4}}, & \mathbf{x}_2 &\in \mathbb{R}^{384 \times \frac{H}{8} \times \frac{W}{8}} \\ \mathbf{x}_3 &\in \mathbb{R}^{256 \times \frac{H}{16} \times \frac{W}{16}}, & \mathbf{x}_4 &\in \mathbb{R}^{768 \times \frac{H}{32} \times \frac{W}{32}}. \end{aligned}$$

At the final stage of the Swin Transformer, we downsample the feature vector $\mathbf{x}_4$ to obtain an additional low-dimensional feature map $\mathbf{x}_5 \in \mathbb{R}^{1536 \times (H/64) \times (W/64)}$. Then, all feature maps are transformed to the same feature channel number 256 (1×1 convolution), forming a new feature set. This feature set contains various signal information extracted from the subspectrogram, including the signal position, amplitude, and duration of the drone flight control and image transmission signals. These features will subsequently be utilized to detect drone signals and differentiate them from interference signals (e.g., Bluetooth and Wi-Fi signals) in DETR.

After positional encoding, the feature set is fed into the multiscale self-attention (MSSA) as feature input ($\mathbf{X}^f$). The following equations show how MSSA extracts multiscale feature maps with the same resolution:

$$\text{MSSA}\left(\left\{\mathbf{x}_l^f\right\}_{l=1}^L\right)=\sum_{m=1}^M \mathbf{W}_m^{\text{MSSA}}\left[\sum_{l=1}^L \mathbf{A}_{m,l}^{\text{MSSA}} \cdot \mathbf{W}_m' \mathbf{x}_l^f\right] \quad (8)$$

$$\mathbf{A}_m^{\text{MSSA}}=\exp \left(\frac{\mathbf{x}_l^f \mathbf{U}_m^T \mathbf{V}_m \mathbf{x}_l^f}{\sqrt{C_v}}\right). \quad (9)$$

Here, M indexes the attention head, and L indexes the input feature level. $\mathbf{A}_m^{\text{MSSA}}$ represents the result of a single self-attention. $\mathbf{x}_l^f$ is the query, and $\mathbf{U}_m$ is the weight for embedding the query in the m th self-attention head. $\mathbf{V}_m$ is the weight for embedding the key, and $\mathbf{x}_l^f$ represents the key. C_v is the length of the key. $\mathbf{W}_m^{\text{MSSA}}$ represents the weights of the FC layer connected after the attention block.

The encoder output is fed into the cross-attention mechanism in the decoder, where it interacts with the object query ($\mathbf{X}^{\text{OQ}}$) after self-attention to obtain the predicted bounding boxes and classes.

Equations (10)–(13) show the detailed calculations in the decoder

$$\text{MSA}\left(\mathbf{X}^{\text{OQ}}\right)=\sum_{m=1}^M \mathbf{W}_m^{\text{MSA}}\left[\sum_{k \in \Omega_k} \mathbf{A}_m^{\text{MSA}} \cdot \mathbf{W}_m' \mathbf{x}_k^{\text{OQ}}\right] \quad (10)$$

$$\mathbf{A}_m^{\text{MSA}}=\exp \left(\frac{\mathbf{x}_k^{\text{OQ}} \mathbf{U}_m^T \mathbf{V}_m \mathbf{x}_k^{\text{OQ}}}{\sqrt{C_v}}\right). \quad (11)$$

Let us refer to $\text{MSA}(\mathbf{x}^{\text{OQ}})$ by $\mathbf{X}^*$ and encoder outputs by $\mathbf{X}^{\text{EC}}$. The following equations show the multihead cross attention (MCA):

$$\text{MCA}\left(\mathbf{X}^*, \mathbf{X}^{\text{EC}}\right)=\sum_{m=1}^M \mathbf{W}_m^{\text{MCA}}\left[\sum_{k \in \Omega_k} \mathbf{A}_m^{\text{MCA}} \cdot \mathbf{W}_m' \mathbf{X}^*\right] \quad (12)$$

$$\mathbf{A}_m^{\text{MCA}}=\exp \left(\frac{\mathbf{x}_k^{\text{EC}} \mathbf{U}_m^T \mathbf{V}_m \mathbf{x}_k^*}{\sqrt{C_v}}\right). \quad (13)$$

Here, $\mathbf{A}_m^{\text{MSA}}$ represents the result of a single cross-attention. $\mathbf{x}_k^*$ is the values and key. $\mathbf{x}_k^{\text{EC}}$ is the query. $\mathbf{W}_m^{\text{MCA}}$ is the weight of the FC layer connected after the cross-attention block.

Through the aforementioned processing, we extract specific information about the drone signal from the subspectrogram using the feature set, including the frequency band and type of the drone signal.

B. Second Stage: UAV Classification Based on ViT

In the field of signal processing, extensive research has been conducted utilizing CNN to perform a variety of classification tasks, including signal modulation style classification. These approaches either employ CNN to classify raw data directly or to classify feature representations derived from the raw data. Building on this body of work, this article leverages the distinctive time–frequency representations of different UAVs to achieve UAV classification.

Traditional CNN-based image classification networks heavily rely on the convolutional layers to extract features from images. While convolution operations can accurately extract pixel-level features, the interconnection between each convolution kernel is relatively weak. This issue is exacerbated in low signal-to-noise ratio (SNR) conditions, where the difference in pixel values between signal and background noise is often minimal. We show the drone signals under different SNR spectrograms in Appendix III.

Therefore, we selected the ViT-B/16 model as our second-stage model. The entire structure of the ViT-B/16 model is shown in Fig. 5.

First, we segment the subwindows $\mathbf{X}_{\text{SW}} \in \mathbb{R}^{3 \times H \times W}$ obtained from the first-stage model into smaller patches $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot 3)}$, where $(H \times W)$ is the resolution of the original image, P is the resolution of the small patches, and $N = (HW/P^2)$ is the number of small patches. Subsequently, we embedded these patches using a layer with hidden units of size D and applied positional encoding, transforming the small patches into input vectors $\mathbf{X}_0 \in \mathbb{R}^{N \times D}$ for the encoder. The following equation utilized W-MSA to compute these feature vectors L times:

$$\mathbf{X}_l = \mathbf{W}_l \cdot \text{LN}[\text{W-MSA}\{\text{LN}(\mathbf{X}_{l-1})\}]. \quad (14)$$

Here, $\mathbf{W}_l$ represents the multilayer perceptron (MLP) weights connected to the L th encoder. LN and W-MSA are the same mechanisms used in Section III-A. The following equations show that the final feature vectors are input into a linear layer to output the probability distribution:

$$\mathbf{X}^{\text{class}} = \mathbf{W}^{\text{Linear}} \cdot \mathbf{X}_L \quad (15)$$

$$o_i = \frac{\exp\left(\mathbf{x}_i^{\text{class}}\right)}{\sum_{n=1}^N \exp\left(\mathbf{x}_n^{\text{class}}\right)}, \quad (i = 1, \dots, N). \quad (16)$$

Here, $\mathbf{X}^{\text{class}}$ represents the output of the classify model in tensor form, and $\mathbf{W}^{\text{Linear}}$ represents the weights of the linear layer in classify head. By applying the softmax function in (16), we transform the inference results from tensor into probability of each class. In (16), N indexes the total number of predicted categories, and o_i represents the probability of each class. By concatenating the probability results,

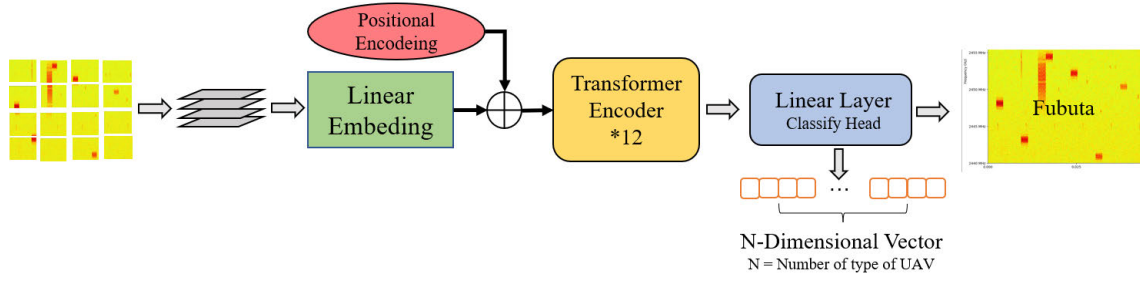


Fig. 5. Structure of the second-stage model (ViT-B/16).

we obtained $\mathbf{O} = [o_1, o_2, \dots, o_i, \dots, o_{n-1}, o_n]$ a probability distribution vector as the model predicted results for each spectrogram. The category associated with the highest prediction probability identified as the type of drone determined by the model.

In the data processing phase, we applied both data cleaning and data augmentation techniques. Data cleaning was performed to reduce the occurrence of extreme or outlier data. Data augmentation was pivotal in enhancing the model's ability to learn drone signal features from time–frequency representations. Specifically, we employed techniques such as Gaussian noise injection, contrast adjustment, and random cropping to diversify the training data and bolster the model's generalization capability.

During the training process, we first conducted pretraining on the ImageNet using the original ViT-B/16 model. Subsequently, we fine-tuned this pretrained model, which was replaced with a new detection head (linear layer) on our specific dataset. Transfer learning alleviates the computational burden for models to comprehend intricate data.

To facilitate the model's convergence on a large dataset, we employed a warm-up training strategy in the initial stages. Specifically, the model was trained with a gradually increasing learning rate starting from a lower value over several epochs, which mitigated potential instability during early training phases. Furthermore, we implemented a cosine annealing learning rate scheduler alongside an increased batch size. This combination enabled the model to dynamically adjust the learning rate according to the training progress, thereby enhancing the effectiveness of fine-tuning.

C. System Detail Explanation

Why Windowing Before First-Stage Model? Commercial drones, like DJI, often have automatic and manual communication frequency selection modes. When the pilot uses the automatic mode, the flight controller selects a frequency band with less noise for communication automatically. In such scenarios, it is necessary to monitor a wide spectrogram band. This results in the model will search signal in a massive spectrogram. Directly feeding such spectrogram into the first-stage model is impossible due to the high compute complexity $O(4hwC^2 + 2h^2w^2C)$. Therefore, we window the entire spectrogram into smaller windows before feeding it into the model in the first stage. This operation standardizes the size of the input windows for the model and reduces the computational complexity to $O(hwC^2) + O((h^2w^2/M^2)C)$. The results of

our experiments demonstrate that by utilizing the 3090 GPU for inference and setting $M = 2$, the average inference time on a 7400×800 resolution spectrogram (1850×800 after windowing) decreases from 376.81 to 243.79 ms (60.9475×4). Consequently, this leads to a notable reduction in processing time of approximately 36% compared to methodologies that do not incorporate windowing techniques. For calculation details, see Appendix II.

Why Window Filtering Before The Second-Stage Model? The window filtering algorithm is a postprocessing step for the first-stage signal detection results. It reduces the number of empty windows passed to the second-stage model, thereby decreasing its processing time. Additionally, it addresses the issue of signal splitting caused by windowing, where a signal might be divided between two adjacent windows. This split can distort the signal shape, especially for wideband signals, leading to inaccurate drone-type recognition. To resolve these issues, the following equations described a new window filtering algorithm:

$$y_{\text{Avg}} = \frac{\sum_{n=1}^N (y_1 + y_2 + \dots + y_n)}{N} \quad (17)$$

$$\text{Coor}_{W1} = (x_W, y_{\text{Avg}}) \quad (18)$$

$$\text{Coor}_{W2} = (x_W, y_{\min}) \quad (19)$$

$$\text{Coor}_{W3} = (x_W, y_{\max}). \quad (20)$$

Here, N is the number of detected FHSS signals in a single subwindow. y_n represents the frequency band (y-axis) of the n th FHSS signal. y_{Avg} is the average frequency point of all FHSS signals within the current window. x_W is the central time slot (x-axis) of the current window. In (18)–(20), we derive extra three predicted window center coordinates, Coor_{W1} , Coor_{W2} , and Coor_{W3} . These coordinates use fixed window lengths (l_W) and heights (h_W) to generate three extra subwindows, including the lower and upper parts of the origin subwindow. The proposed design ensures that at least one of the generated windows encompasses the entire drone signal, thereby mitigating the miss rate.

IV. EXPERIMENTS

A. Dataset

We create an innovative dataset based on raw data collected by DroneRFa [33]. The DroneRFa contains 22 types of RF datasets, encompassing 16 different drone types and six different types of controllers. The drone types include Air2S, AVATA, DJI self-assembled drone,

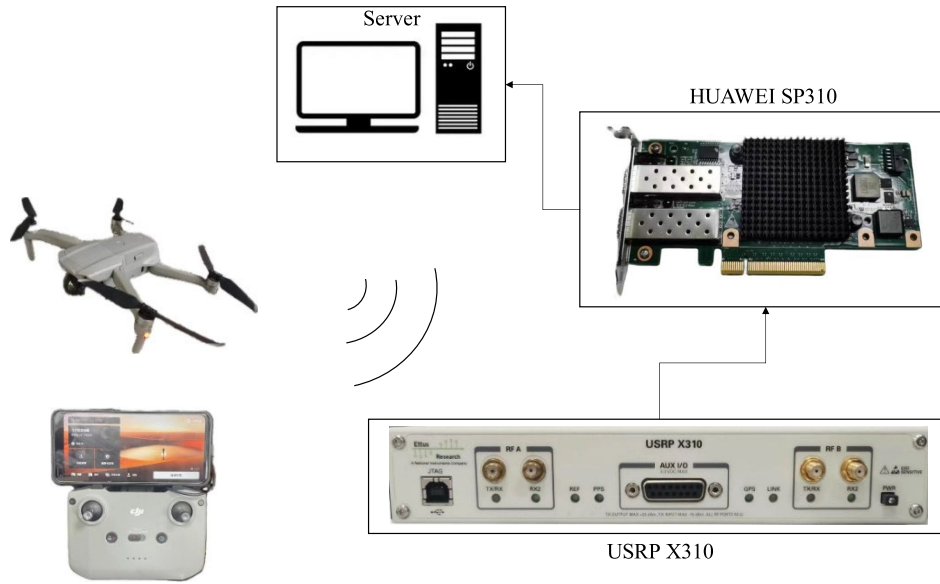


Fig. 6. Hardware connection of experimental platform.

Inspire2, Matrice30T, Matrice100, Matrice200, Matrice300, Matrice600Pro, Mavic3, MavicPro, Mini2, Mini3Pro, Phantom3, Phantom4Pro, and Phantom4ProRTK; and the controller types include FutabaT14SG, FutabaT61Z, RadioLinkAT9S, TarainsX90Plus, Vbar, and YunzhuoT12. The signal clip with a duration of 0.1 s and a bandwidth of 100 MHz is subjected to STFT using the Hamming window and 2048 STFT points. The STFT results are subsequently transformed into spectrograms using the jet color mapping rule. These spectrograms are employed to build an object detection dataset for training improved DETR and an image classification dataset for fine-tuning the ViT-B/16.

In the object detection dataset, signals are categorized into 15 distinct classes based on their varying durations and bandwidths. Reaching a finer distinction is unattainable due to the presence of signals with analogous characteristics. The training data for the first-stage model comprise spectrogram clips that are annotated with occurrence time, frequency, and specific types of drone signals. We allocate 10% of the total spectrograms with labels as the training set. The remaining images constitute the test set, serving as a comprehensive evaluation benchmark.

In the classification dataset, we divide the spectrogram into 22 categories. The data utilized to fine-tune the second-stage model consist of spectrogram clips labeled with drone type. We partition the total images into two sets, with 50% of the images allocated as the training set and the remaining images designated as the test set.

B. Setup and Test

The RFDroneViT is implemented using Python, while the improved DETR is implemented based on the Openmmlab [34] framework utilizing PyTorch. The ViT-B/16 is implemented based on the HuggingFace framework using PyTorch.

The RFDroneViT is implemented on a computer system featuring an AMD 5900X CPU and Nvidia 3090 GPU. The

hardware connection used in this experiment is shown in Fig. 6. The HUAWEI SP310 serves as the intermediary connecting the host computer and the receiver.

The USRPX310, functioning as the data receiver, is operated through the utilization of GNURadio, an open-source software platform, thereby constituting a comprehensive SDR system. The receiving process is conducted in a sequential manner, where one second of raw data is collected and stored, followed by the subsequent one-second collection period. This receive mode ensures a fixed size for each collected data packet, thereby standardizing the shape of the windowing algorithm and the number of input windows in the first stage.

To validate the system's reliability, we conduct a practical experiment. During the experiment, we conduct system deployment in an environment with concurrent signal interference. Subsequently, we gradually moved both the pilot and the drone away from the system until the receiving device ceased to capture the RF signal emitted by the drone. Throughout this process, we record and preserve all RF signals generated by the drone. The RF signals generated during the process are input into the system for verification, and the corresponding verification results are shown in Fig. 7.

Fig. 7 illustrates the system's remarkable capability to accurately detect and identify the target drone within the zone where the USRPX310 can collect signals. The first-stage model captures every target signal in a spectrogram with a certain frequency band and time that the drone signal appears. The second-stage model identifies the drone type appearing at a certain time and frequency based on the detection result.

C. Comparison With Other Methods

In this section, we primarily present the performance comparison results of our designed models with other mainstream models of the same type in our dataset.

To verify the superior performance of our model among the other models, we conduct a comparison experiment on the first-stage model. For the sake of fairness, we use the

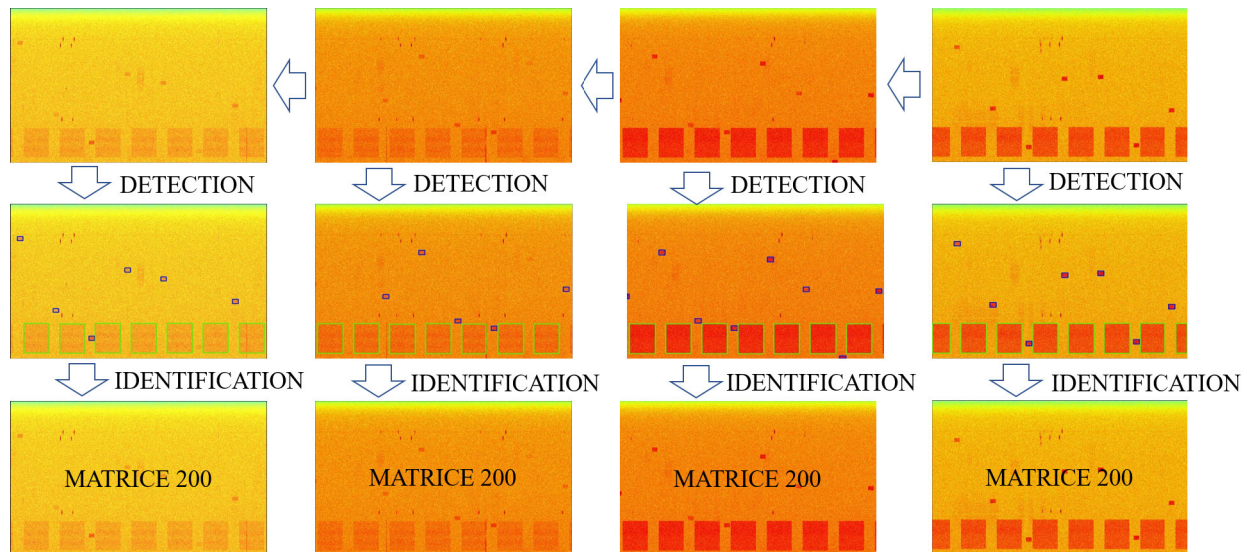


Fig. 7. Results of the experiments of Matrice200. To concisely represent the results of the drone detection, the bounding box associated classifications and predicted probabilities are concealed.

best-performance model on the training set to evaluate the accuracy of a unified validation set.

The comparative outcomes of the first-stage model are presented in Table I. The evaluation metrics we used include standard average precision (AP), AP_{75} (AP IoU > 0.75), AP_{50} (AP IoU > 0.50), AP_S (AP for small targets), AP_M (AP for medium targets), and AP_L (AP for large targets). The models included in this comparison are representative object detection networks such as YOLOv5, YOLOv8 [35], Improved YOLOv8 (replacing origin backbone with Swin Transformer), Faster-RCNN (using FPN) with ResNet50 backbone and ResNet101 backbone, Faster-RCNN with ResNet50 backbone and ResNet101 backbone, and other DETR-like model include Deformable-DETR, DAB-DETR, and DINO.

As shown in Table I, DETR with the large Swin Transformer as the backbone yields an improvement of +9.2 AP compared with the best performance Faster-RCNN model. Similarly, DETR with a tiny Swin Transformer backbone yields an improvement of +7.3 AP. The significant enhancement in AP attests to the substantial advancements achieved by our model when juxtaposed with the conventional mainstream. Compared with the best-performance YOLO model, DETR with the large Swin Transformer backbone yields an improvement of +5.4 AP, while DETR with the tiny Swin Transformer exhibits an enhancement of +3.5 AP. DETR with the tiny Swin Transformer backbone outperforms DAB-DETR with ResNet50 backbone by +4.5 AP, outperforms Deformable-DETR with ResNet50 backbone by +3.8 AP, and surpasses DINO-5scale with ResNet50 backbone by +2.0 AP. Furthermore, incorporating the large Swin Transformer backbone into DETR leads to a substantial performance enhancement of +6.4 AP compared to Deformable-DETR with ResNet50 backbone, +5.7 AP compared to DAB-DETR with ResNet50 backbone, and +3.9 AP compared to DINO-5scale with ResNet50 backbone.

Notably, compared with the best-performance model without using FPN, DETR with the tiny Swin Transformer

TABLE II
RESULT OF THE SECOND-STAGE MODEL COMPARISON

Model	dataset	Top-1(%)
ResNet50	ImageNet	64.8
ResNet50	Drone Classification	90.2
ResNet101	ImageNet	70.8
ResNet101	Drone Classification	91.3
MobileNetV2-S [36]	ImageNet	72.1
MobileNetV2-S	Drone Classification	92.1
MobileNetV2-L	ImageNet	74.7
MobileNetV2-L	Drone Classification	93.8
EfficientNetV2-S [37]	ImageNet	83.9
EfficientNetV2-S	Drone Classification	96.4
EfficientNetV2-M	ImageNet	85.1
EfficientNetV2-M	Drone Classification	97.1
EfficientNetV2-L	ImageNet	85.7
EfficientNetV2-L	Drone Classification	97.5
ViT-B/16	ImageNet	88.6
ViT-B/16	Drone Classification	98.7

backbone yields a significant improvement of +9.7 APs, while DETR with the large Swin Transformer backbone exhibits an even more remarkable enhancement of +13.7 APs.

We further conduct another comparative experiment on the second-stage model to validate the superior performance of our model compared to other models. The evaluation metric used in this experiment is standard Top-1 accuracy.

During training, we use transfer learning technology to accelerate the convergence of the model on the dataset. We initially conduct pretraining of the model using the publicly available ImageNet dataset, followed by fine-tuning the optimal pretraining outcome on our specific training set. The model's performance on both ImageNet and our validation set is presented in Table II.

As shown in Table II, ViT-16/B yields +23.8% Top-1 on ImageNet compared with ResNet50 and +17.8% Top-1 on ImageNet compared with ResNet101. On our dataset, ViT-16/B yields +8.5% Top-1 compared with ResNet50 and +7.4% Top-1 compared with ResNet101. ViT-16/B yields +16.5% Top-1 on ImageNet compared with MobileNetV2-S,

TABLE III
RESULT OF BACKBONE REPLACEMENT

Backbone Type	AP	AP ₇₅	AP ₅₀	AP _S	AP _M	AP _L
ResNet50+FPN	70.2	80.9	94.1	56.7	66.9	79.9
ResNet101+FPN	71.6	81.5	94.9	57.2	68.7	80.1
Swin-T+FPN	72.4	82.1	95.9	58.9	69.7	81.8
Swin-B+FPN	73.9	83.1	96.4	60.4	70.4	82.2
Swin-L+FPN	74.3	84.6	98.9	62.9	71.8	84.8

TABLE IV
COMPARISON OF SWIN-L WITH/WITHOUT FPN

Methods	AP	AP ₇₅	AP ₅₀	AP _S	AP _M	AP _L
Swin-T(normal)	70.8	80.7	95.1	53.6	68.9	82.1
Swin-T(FPN)	72.4	82.1	95.9	58.9	69.7	82.2
Swin-B(normal)	71.1	81.3	95.9	53.9	69.9	82.9
Swin-B(FPN)	73.9	83.1	96.4	60.4	70.4	83.3
Swin-L(normal)	72.6	81.9	94.1	54.2	70.1	83.7
Swin-L(FPN)	74.3	84.6	98.9	62.9	71.8	84.8

and +13.9% Top-1 on ImageNet compared with MobileNetV2-L. On our dataset, ViT-16/B yields +6.6% Top-1 compared with MobileNetV2-S and +4.9% Top-1 compared with MobileNetV2-L. ViT-16/B yields +4.7% Top-1 on ImageNet compared with EfficientNetV2-S, +3.5% Top-1 on ImageNet compared with EfficientNetV2-M, and +2.9% Top-1 compared with EfficientNetV2-L. On our dataset, ViT-16/B yields +2.3% Top-1 compared with EfficientNetV2-S, +1.6% Top-1 compared with EfficientNetV2-M, and +1.2% Top-1 compared with EfficientNetV2-L.

The result of the comparison of the second-stage model suggests that our model demonstrates superior accuracy Top-1.

D. Ablation Study

To assess the enhancement in the detection performance of the first-stage model resulting from the incorporation of new modules, we conduct ablation experiments in this section.

The comparative outcomes of the enhancements made to the backbone network are presented in Table III. The evaluation metrics include standard AP, AP₇₅, AP₅₀, AP_S, AP_M, and AP_L, which align with the metrics presented in Table I. The comparison objects are mainly two mainstream feature extraction networks: Swin Transformer and ResNet. We employ the FPN method in both types of backbone architectures and utilize identical training and validation datasets.

As shown in Table III, compared with ResNet50, our method yields improvements of +4.1 AP and +2.2 AP with tiny Swin Transformer. It is worth noting that the incorporation of Transformer techniques in a tiny Swin Transformer still yields an enhancement of +0.8 AP, compared with the best CNN-based backbone, ResNet101.

We employ the identical approach to perform ablation experiments on the enhancements made to FPN and present the corresponding experimental results in Table IV.

As shown in Table IV, the FPN method results in performance improvements of +5.3 AP_S for the tiny Swin Transformer backbone, +6.5 AP_S for the based Swin Transformer, and +8.7 AP_S for the large Swin Transformer. The

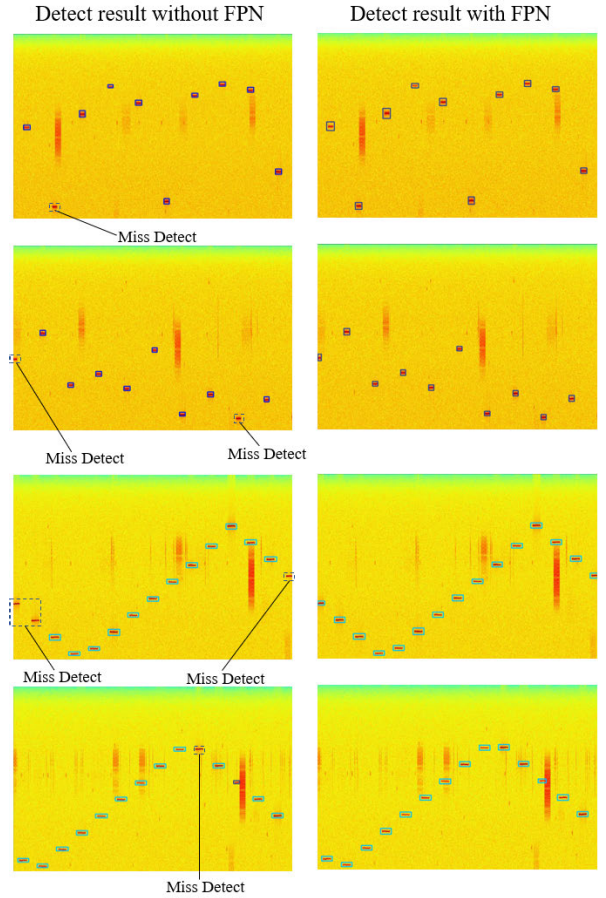


Fig. 8. Two types of drone signal detection clips were obtained without/with the FPN model. The drone on the upside clips is Vbar, and the one downside clip is FrskyX20.

significant enhancement in AP_S demonstrates that the model after introducing the FPN method has a notable improvement in accuracy on small target detection, as compared to the original model. For further elaboration, the visual outcome of FPN enhancement is shown in Fig. 8. For this system, the significant improvement in small object detection brought by FPN is conducive to reducing the missed detection rate and the classification error rate of UAVs.

V. LIMITATIONS AND FUTURE OPPORTUNITIES

Considering such a scenario, we are monitoring a spectrogram with a bandwidth of 100 MHz to detect the simultaneous presence of two drones operating on the 2.4-GHz-frequency channel. In these situations, the efficacy of RFDroneViT is significantly compromised, as depicted in Fig. 9.

From the perspective of drone behavior, such situations are infrequent. Modern drone controllers typically select channels with minimal interference when establishing communication links, thereby operating on relatively clean spectrum bands. However, it is noteworthy that frequency spectrum overlap can occur during the connection phase between the drone and the remote controller (take-off stage).

From the perspective of deep learning, this scenario can be classified as a heavily occluded target object case. However, viable solutions exist to address this challenge. As first-stage

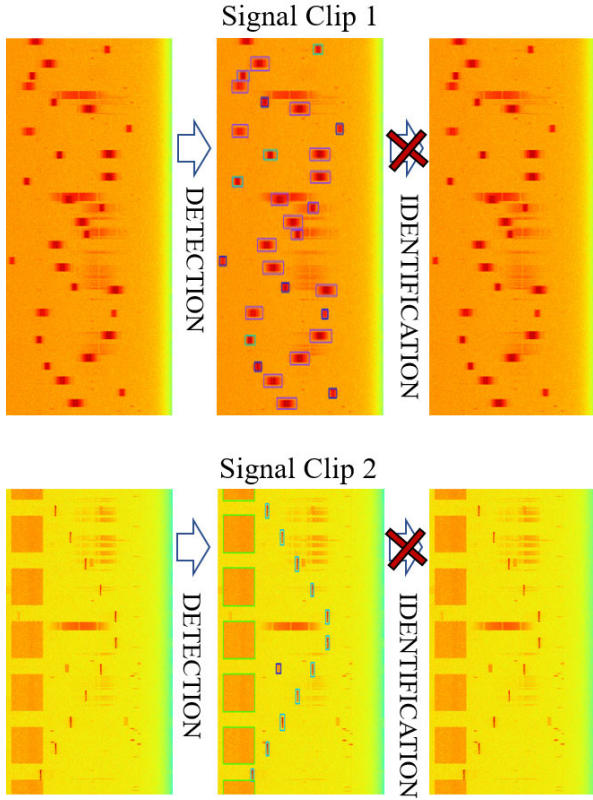


Fig. 9. Two signal clips, each featuring simultaneous appearances of two different drones. Clip 1 includes the Futaba T14SG and RadioLink AT9S, while Clip 2 features the Frsky X20 and Matrice200.

models remain unaffected by such interference, drones in the specified frequency bands can still be accurately detected. Prior to proceeding with the second stage, we can employ a signal processing technique called FH signal sorting [38]. This technique enables the analysis of detected radiation sources and differentiation between signals originating from distinct sources. By performing signal separation, the processed signals closely resemble those present in the dataset, facilitating the achievement of comparable technical outcomes.

Based on the aforementioned analysis, our system exhibits a limitation when multiple drones simultaneously take off within the designated detection area. In future research endeavors, we will delve deeper into this technique and continuously enhance the feasibility of the proposed solutions.

VI. CONCLUSION

This article integrates the ViT into the domain of drone detection and classification, devising a novel network architecture that constitutes a fully operational system. The improved DETR achieves the state-of-the-art performance in drone detection, achieving an impressive AP of 74.3. Meanwhile, the ViT-B/16 model demonstrates exceptional accuracy in drone identification, attaining a remarkable Top-1 accuracy rate of 98.7%. Our architectural improvements have proven to significantly enhance the system's performance. The integration of the FPN into the DETR, as proposed in this study, can be prospectively adapted for enhanced small object

detection across a wider array of scenarios. Furthermore, the spectrogram windowing and spectrogram window filtering algorithms proposed in this article are adaptable for application in a broader spectrum of spectrogram-based sensing signal processing tasks. Moreover, the proposed methodology for drone detection and identification is adaptable to an expanded array of deep learning-oriented signal detection and modulation recognition challenges. However, this system still has shortcomings that the recognition results will have a delay if there have multiple drones taking off in the detection area at the same time.

APPENDIX I

SPECTROGRAM FROM DIFFERENT UAVs

In this section, we show the spectrograms of different drones and controllers. The original signal clips with a duration of 0.1 s and bandwidth of 100 MHz, and the STFT results were mapped to color using the jet rule. As shown in Fig. 10, we can see that the spectrograms from different UAVs have great differences in vision, and even the individual image transmission or flight control signals have certain differences.

APPENDIX II

COMPUTATIONAL COMPLEXITY FOR WINDOWING

In this section, we demonstrate how the spectrogram windowing operation can reduce computational complexity. To provide a clear comparison, we first calculate the computational complexity of the original multihead self-attention (MSA) mechanism in the Swin Transformer. First assume the following.

- 1) The height and width of the input feature map are h and w , respectively.
- 2) The number of channels is C .
- 3) The number of positions (tokens) is $N = hw$.

Equations (21)–(24) are the overall computational complexity of MSA.

First, we should perform the *linear transformations*. Before the self-attention, the input features are linearly transformed to generate the query (Q), key (K), and value (V). Each linear transformation has a complexity of

$$\text{Linear Transform: } O(hwC \times C) = O(hwC^2). \quad (21)$$

Since there are three linear transformations (one for each of Q , K , and V), the total complexity is $O(3hwC^2)$. Additionally, there is a final linear transformation after concatenating the multihead outputs $O(hwC^2)$. Thus, the total complexity for linear transformations is

$$O(3hwC^2) + O(hwC^2) = O(4hwC^2). \quad (22)$$

Then, we perform *attention weights and weighted sum*. For each head, the dot product between the Q and K needs to be computed. Assuming there are M heads, and each head has a dimension $d_k = C/M$, the complexity of computing QK^T is

$$O(hw \times hw \times d_k) = O\left(h^2w^2 \times \frac{C}{M}\right). \quad (23)$$

Since there are M heads, the total complexity is $O(h^2w^2C)$.

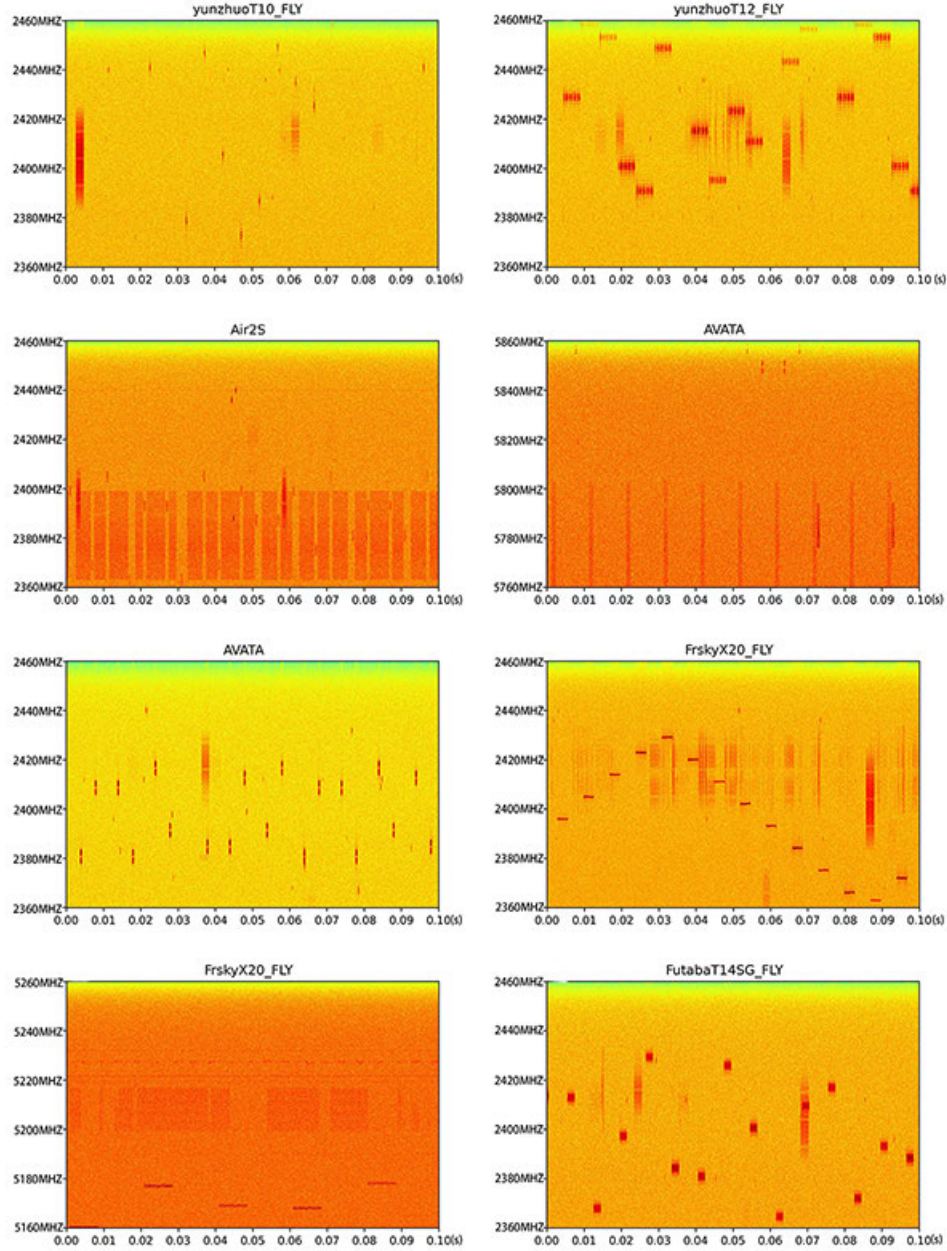


Fig. 10. Results of the visualized STFT by using jet color mapping rules.

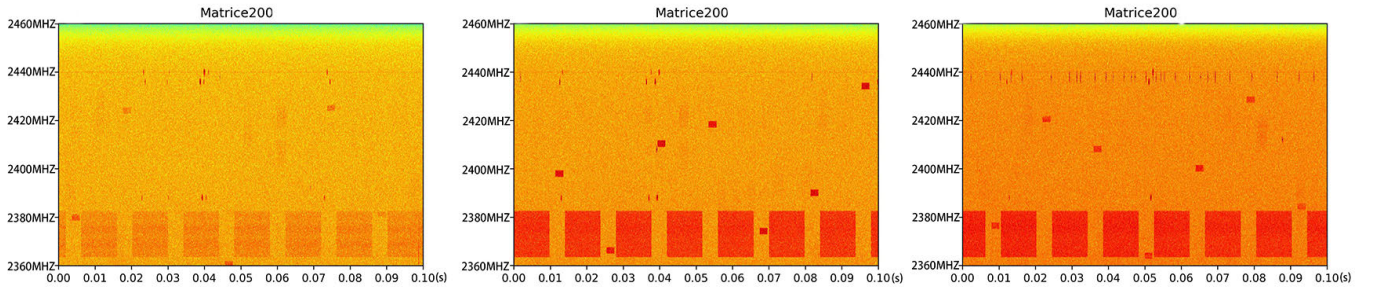


Fig. 11. Spectrograms of Matrice200 under different SNRs.

Next, the attention weights are used to compute a weighted sum with the V

$$O(hw \times hw \times d_k) = O\left(h^2 w^2 \times \frac{C}{M}\right). \quad (24)$$

Again, for M heads, the total complexity is $O(h^2 w^2 C)$. Therefore, the total complexity for computing attention weights and weighted sums is $O(2h^2 w^2 C)$.

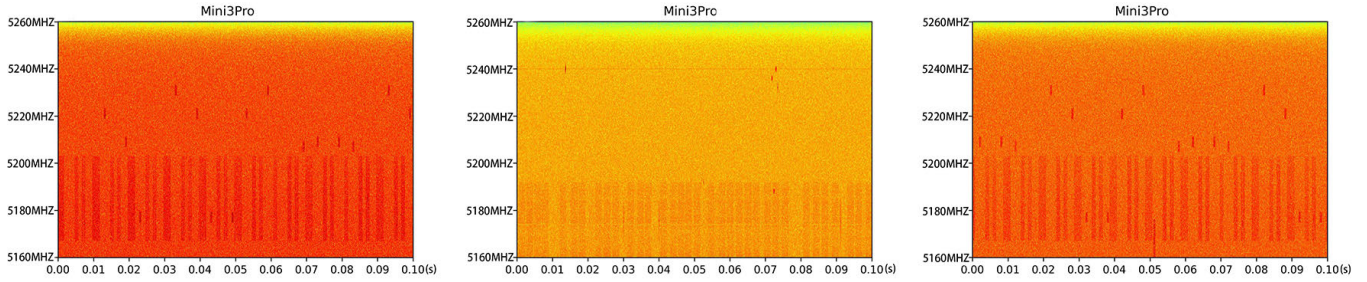


Fig. 12. Spectrograms of Mini3Pro under different SNRs.

Finally, the total computational complexity in the original MSA is $O[4hwC^2 + 2(hw)^2C]$.

If we divide the original image into M^2 subwindows—each of size h/M by w/M —then the linear transformations retain the same computational complexity as before. However, the computational complexity of attention weights and the weighted sum for each patch is significantly reduced to

$$O\left(M^2 \cdot \frac{h^2w^2}{M^4}C\right) = O\left(\frac{h^2w^2}{M^2}C\right)$$

so the total computational complexity is

$$O(hwC^2) + O\left(\frac{h^2w^2}{M^2}C\right).$$

By splitting the image into M^2 patches and processing each patch separately, the overall computational complexity of the MSA mechanism is reduced. Specifically, the complexity for attention weights and weighted sums is reduced by a factor of M^2 , making it more efficient for larger images.

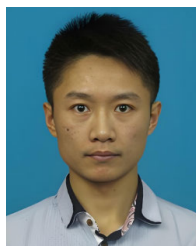
APPENDIX III SPECTROGRAM OF THE SAME UAV UNDER DIFFERENT SNRS

In this section, we show the spectrograms of the same UAV under different SNR conditions in Figs. 11 and 12. As shown in the images, the SNR affects the color brightness of the signal region relative to the background region in the spectrogram.

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